

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1.

Figure 1.1

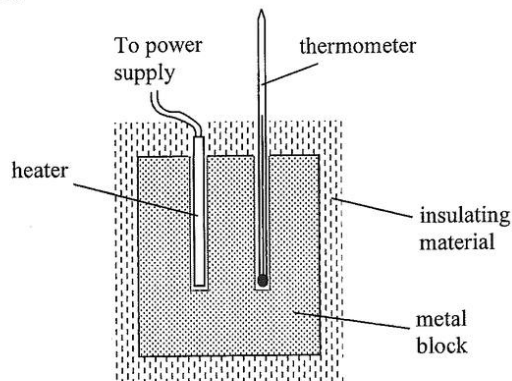
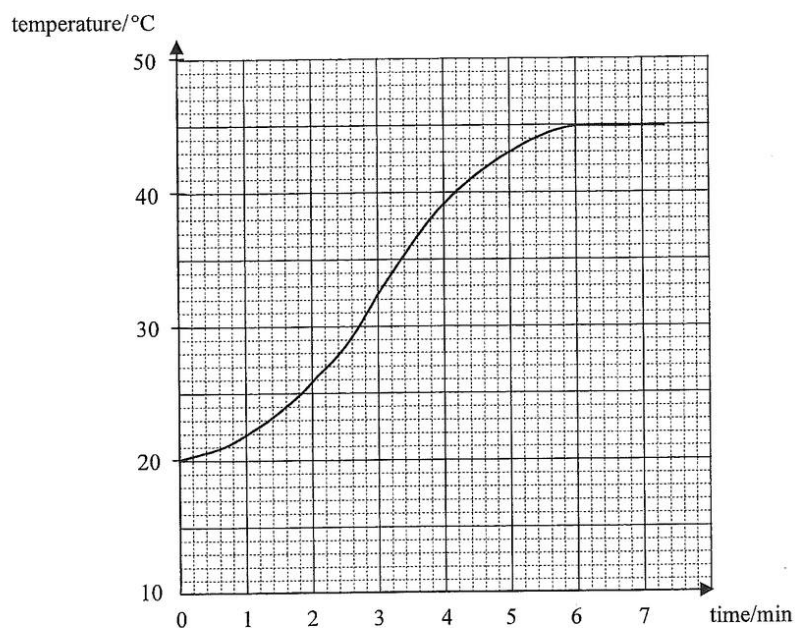


Figure 1.1 shows an experimental set-up to find the specific heat capacity of a metal. The metal block is wrapped by insulating material. A heater is connected to a power supply. It is switched on when the temperature of the metal block is 20°C and then switched off when the temperature reaches 43°C . The graph below shows the variation of the temperature of the metal block with time.



(a) Use the graph to find the duration that the heater is switched on. (1 mark)

(b) After the heater is switched off, the temperature of the metal block continues to rise for a while. Explain why. (1 mark)

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- (c) Given: mass of the metal block = 0.80 kg
heater voltage = 12 V
heater current = 4.0 A

- (i) By considering the maximum temperature rise of the metal block, calculate the specific heat capacity of the metal as found from the experiment. (2 marks)

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- (ii) Would your result be the same, higher, or lower than the actual value of the specific heat capacity of the metal? Explain. (2 marks)

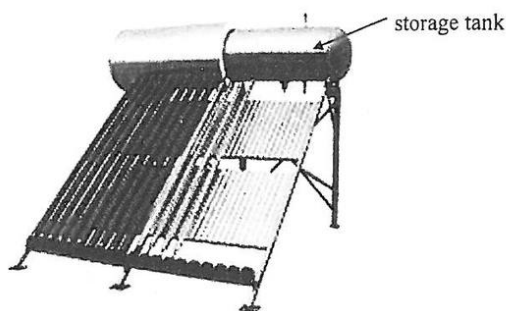
Answers written in the margins will not be marked.

- (d) This method is not suitable for measuring the specific heat capacity of a glass block. Explain. (1 mark)

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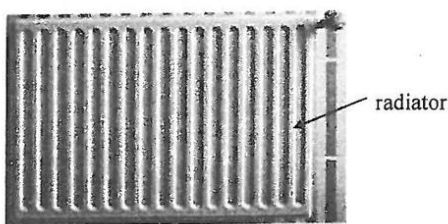
1.

Figure 1.1



A solar water heater shown in Figure 1.1 is installed on the rooftop of a house. During the day, the heater heats up 1.5 m^3 of water to 80°C . At night, the hot water in the storage tank is circulated to the radiators (see Figure 1.2) in different rooms of the house to keep the rooms warm.

Figure 1.2



Given: density of water = 1000 kg m^{-3}
specific heat capacity of water = $4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

- (a) Given that 15% of the energy is lost during the transfer of water, how much heat can be released from the system to the rooms when the water temperature drops to 60°C ? (3 marks)

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- (b) Given that during night time the radiators maintain an average output power of 4.5 kW, how long can the radiators maintain this average power until the water temperature in the system drops to 60 °C ? Give your answer in hours. (2 marks)

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- (c) The rate of heat released by the solar water heating system during the time period calculated in (b) is in fact not constant and gradually drops. Explain why this is so. (1 mark)

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Book 1 section 2.2: DSE long questions

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